

Predicting 30-Day Hospital Readmission in Medicare Patients

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Motivation & Background

- **30-day readmissions** are a major healthcare challenge, linked to high costs and poorer outcomes.
- **Medicare beneficiaries** (age 65+) are particularly vulnerable to readmissions.
- **Reducing readmissions** is a key national priority, reinforced by CMS through the Hospital Readmissions Reduction Program (HRRP).
- **Data-driven prediction** at discharge can help primary care providers identify at-risk patients early and enable timely interventions.

\$26B+

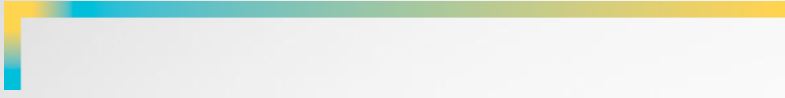
Annual cost of
unplanned readmissions

Problem Definition

- **Challenge:** Identifying Medicare patients at highest risk of 30-day readmission is difficult at the time of discharge.
- **Goal:** Develop a predictive model using MIMIC-IV v3.1 dataset to assess readmission risk for Medicare beneficiaries.
- **Outcome:** Deliver an interpretable risk-scoring tool that supports primary care providers with targeted interventions.
- **Impact:** Missed risk predictions → avoidable hospitalizations, higher costs, poorer patient outcomes.

Research Questions

- **RQ1:** Which **features** are most **predictive** of **30-day readmissions** in Medicare beneficiaries?
- **RQ2:** Which **modeling approach** (statistical, boosting, deep learning, or ensemble) achieves the **best predictive performance**?
- **RQ3:** Can **interpretable ML methods** (e.g., SHAP) provide **actionable insights** for **primary care providers** at discharge?



Dataset

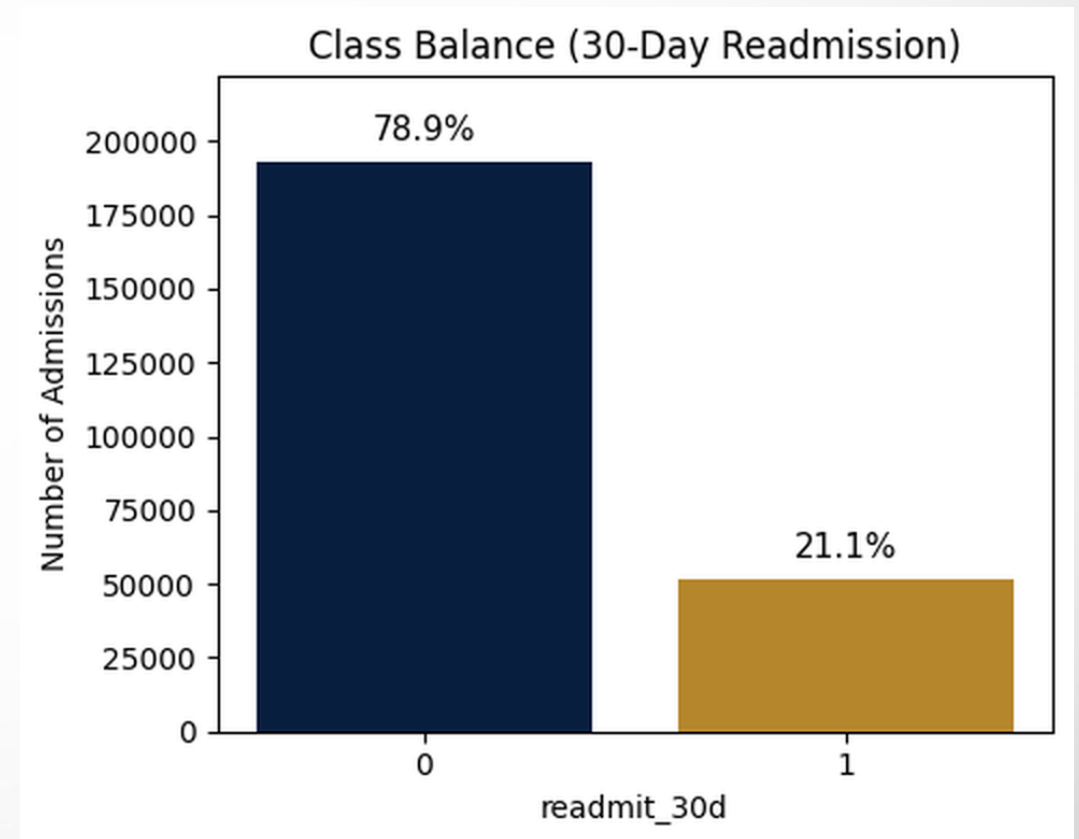
- **Source:** MIMIC-IV v3.1 (PhysioNet) from Beth Israel Deaconess Medical Center, Boston, MA.
- **Full Dataset:** 546,028 admissions with 305 raw features.
- **Cohort:** 244,576 Medicare admissions (insurance == Medicare)
- **Target:** readmit_30d (binary) (21.1% positive rate)

Key Clinical Modules

- Admissions
- Diagnoses (ICD-9/10)
- Procedures
- DRG Codes
- Prescriptions
- Lab Events
- Vital Signs
- ICU Stays
- Transfers
- Microbiology

Exploratory Data Analysis (EDA)

- Dataset is imbalanced: 21.1% of admissions were readmitted within 30 days
- Imbalance justifies using **AUC-ROC** and calibrated probabilities instead of accuracy
- Feature distributions (age, Length of Stay (LOS), comorbidities, meds) show broad clinical variability
- No strong multicollinearity detected among baseline features (not shown for simplicity)



Exploratory Data Analysis (EDA)

Discharge Destination

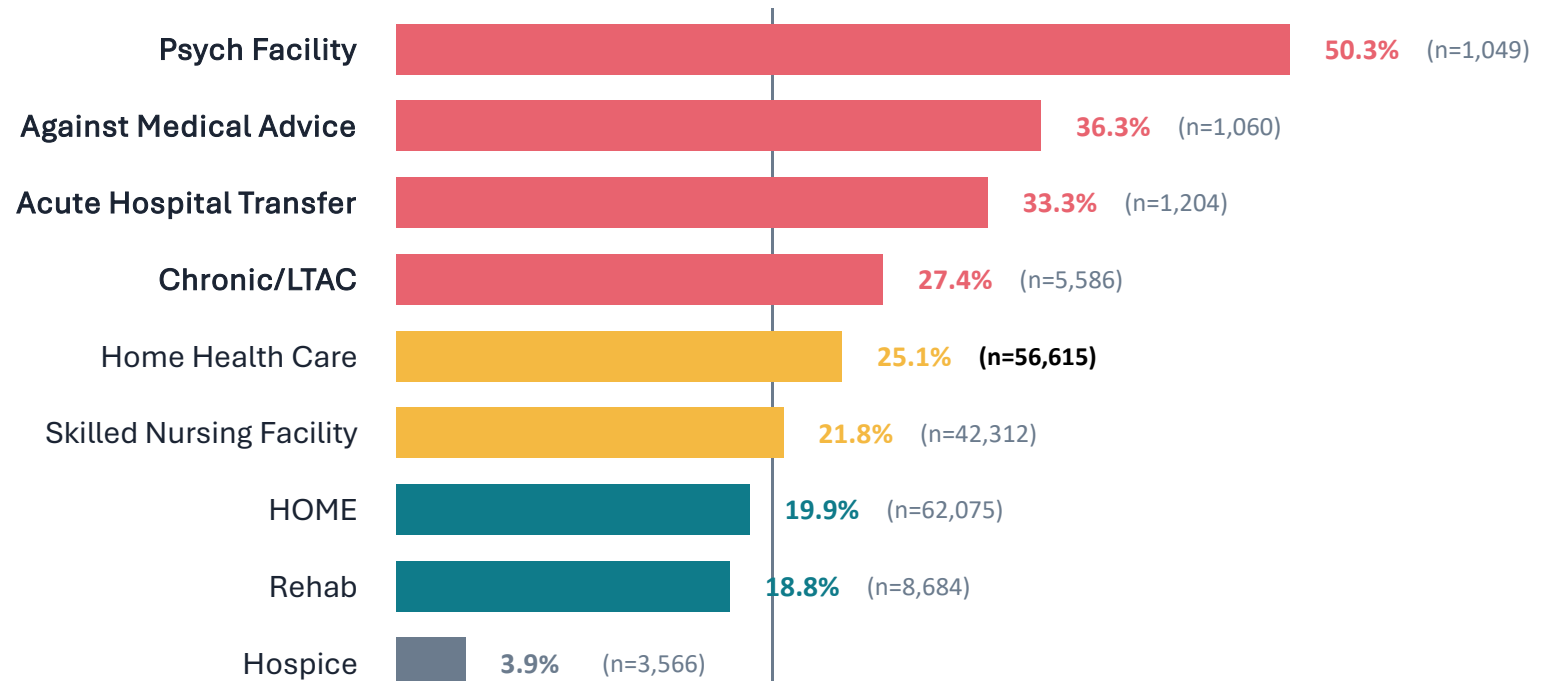
➤ Takeaways:

1. Discharge destination shows wide variation in readmission rates (3.9% to 50.3%), suggesting strong predictive potential.

2. Care-coordination interventions can target the 56K Home Health patients readmitting at 25.1%.

Readmission Rate by Discharge Destination

N = number of admissions | Overall cohort rate: 21.1%



21.1%
cohort avg

Chronic/LTAC = Long-Term Acute Care facilities

Dataset Versions V1-V7

V1: 21
Features

V2: 24
Features

V3: 28
Features

V4: 33
Features

V5: 36
Features

V6: 40
Features

V7: 50
Features

Dataset Versions V1-V6

Version 1 – Baseline Feature Set:

- **Demographics:** age, gender
- **Admission info:** admission type, admission location, discharge location
- **Comorbidities (7 flags):** MI, CHF, diabetes, COPD, cerebrovascular, renal, cancer
- **Stay context:** Length of Stay (LOS), discharge hour, weekend discharge
- **Medication burden:** distinct drugs, polypharmacy flag
- **Prior use (6 months):** prior admissions, mean prior LOS, any ED
- **Diagnosis Related Group (DRG) code**

Table 1: Feature count per model version, showing progressive enrichment from V1 to V6.

Version	Feature Count
V1	21
V2	24
V3	28
V4	33
V5	36
V6	40

Dataset Versions V1-V6

Version 2 – Temporal & Complexity Signals added:

Added 3 engineered features on top of V1:

- **Prior DRG + discharge disposition:** combined category (`last_drg_dispo`) from the previous admission
- **Medication complexity (last 90 days):** Shannon entropy of medication classes (`med_entropy_90d`)
- **Length of Stay (LOS) trend across admissions:** mean change in length of stay vs the prior admission (`los_trend_180d`)

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Dataset Versions V1-V6

Version 3 – Data Quality & Length of Stay Refinement:

Adds 4 engineered features on top of V2:

- **Recomputed Length of Stay (los_days)** – calculated directly as $(discharge - admit) \div 24$ to ensure consistent LOS across all encounters
- **Diagnosis Related Group (DRG) missing flag (drg_code_is_missing)** – 1 if DRG code is null, capturing unusual billing or incomplete documentation
- **Discharge missing flag (discharge_location_is_missing)** – 1 when discharge disposition is absent, signaling atypical workflows
- **Length of Stay missing flag (los_days_is_missing)** – identifies encounters with incomplete timestamps or irregular LOS patterns

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Dataset Versions V1-V6

Version 4 – Clinical Severity & Interaction Features

Added 5 engineered features on top of V3:

- **Age buckets (`age_bucket`)**
 - Age grouped into 4 clinical bands (*lt65, 65–74, 75–84, 85+*).
- **Comorbidity score (`cci_total`)**
 - Sum of all CCI flags to create a single measure of comorbidity burden.
- **Admission acuity score (`admission_type_ord`)**
 - Admission type mapped to an ordered severity scale (ELECTIVE=0 → TRAUMA=3).
- **Age × comorbidity (`age_cci_interaction`)**
 - Interaction term capturing patients who are both older and highly comorbid.
- **Length of Stay × comorbidity (`los_cci_interaction`)**
 - Interaction term capturing long stays combined with higher comorbidity.

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Dataset Versions V1-V6

Version 5 – Nonlinear & Interaction Enhancements

Added 3 engineered features on top of V4:

- **Length of Stay × Age (`los_age_interaction`)**
 - Engineered feature: `los_days * age_at_admit`
 - Captures nonlinear risk for older patients with long stays.
- **Squared comorbidity score (`cci_total_sq`)**
 - Engineered feature: `cci_total ** 2`
 - Amplifies risk for very high comorbidity burden.
- **Log-transformed Length of Stay (`log_los_days`)**
 - Engineered feature: `log1p(los_days)`
 - Reduces effect of extreme LOS values and stabilizes long-stay distributions.

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Dataset Versions V1-V6

Version 6 — Clinical Feature Expansion

In V6, I expanded the dataset by adding **raw clinical information** from other MIMIC-IV tables:

Added Clinical Features (Aggregated per admission):

- **Diagnosis complexity:** `n_diagnoses`
- **Procedure complexity:** `n_procedures`
- **Medication burden:** `n_meds_total`, `n_meds_unique`
- **Lab intensity:** `n_labs_total`, `n_lab_item_types`, `n_labs_abnormal`
- **ICU utilization:** `icu_flag`, `icu_total_hrs`
- **Patient history:** `prior_admissions_all`

Table 1: Feature count per model version, showing progressive enrichment from V1 to V6.

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Dataset Version V7

V6 established 40 core clinical features. V7 adds 10 advanced features in two categories to capture non-linear relationships the base features miss.

Target Encodings — 5 new features

Converts high-cardinality categoricals into their mean readmission rate per category (with K-fold encoding to prevent leakage).

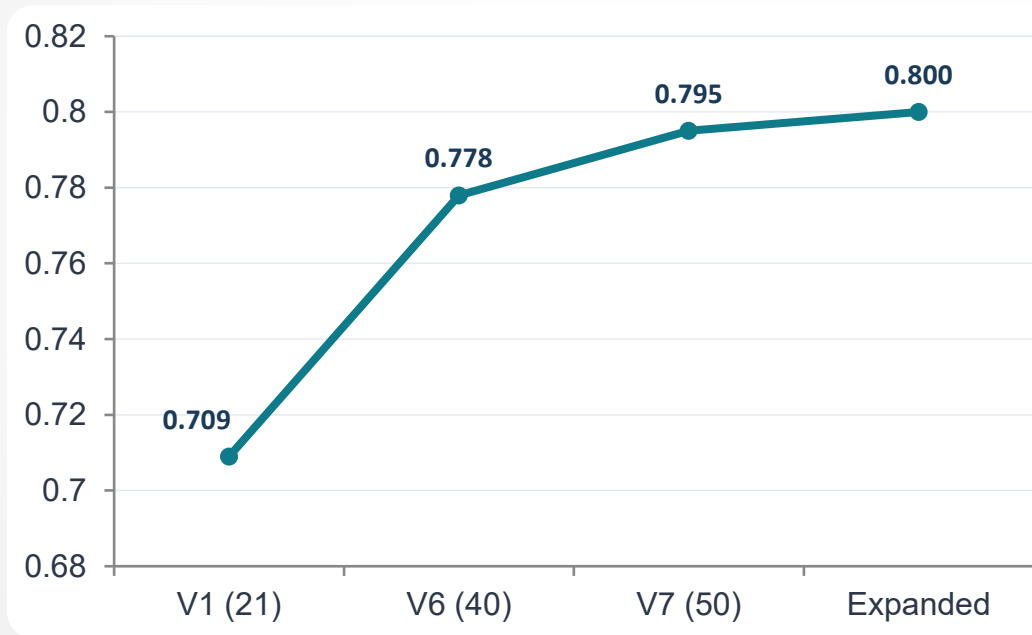
drg_code_te	DRG group → mean readmission rate
discharge_location_te	Discharge destination → readmission rate
primary_dx_chapter_te	ICD chapter → readmission rate
last_drg_dispo_te	Prior DRG + disposition pair → rate
race_te	Race/ethnicity → readmission rate

Clinical Interactions — 5 new features

Multiplies two related clinical features together to capture compound risk that neither feature captures alone.

prior_admits_x_age	Older + frequent admissions = compound risk
renal_risk_x_age	Elderly renal patients at highest risk
severity_x_readmit	Sicker patients with prior readmits
severity_x_lab_abnormal	High severity + abnormal labs
los_trend_x_prior_6m	Rising LOS + frequent recent visits

Feature Expansion V7



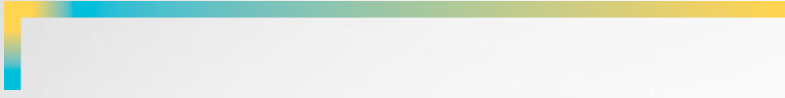
V1 → V7

+0.086 AUROC gain
Core clinical features

V7 → Expanded

+0.005 AUROC gain
Minimal improvement

A core set of ~50 clinically-motivated features captures 99.4% of the predictive signal. Additional feature engineering yields only marginal improvement, confirming the parsimonious model as ideal for deployment.

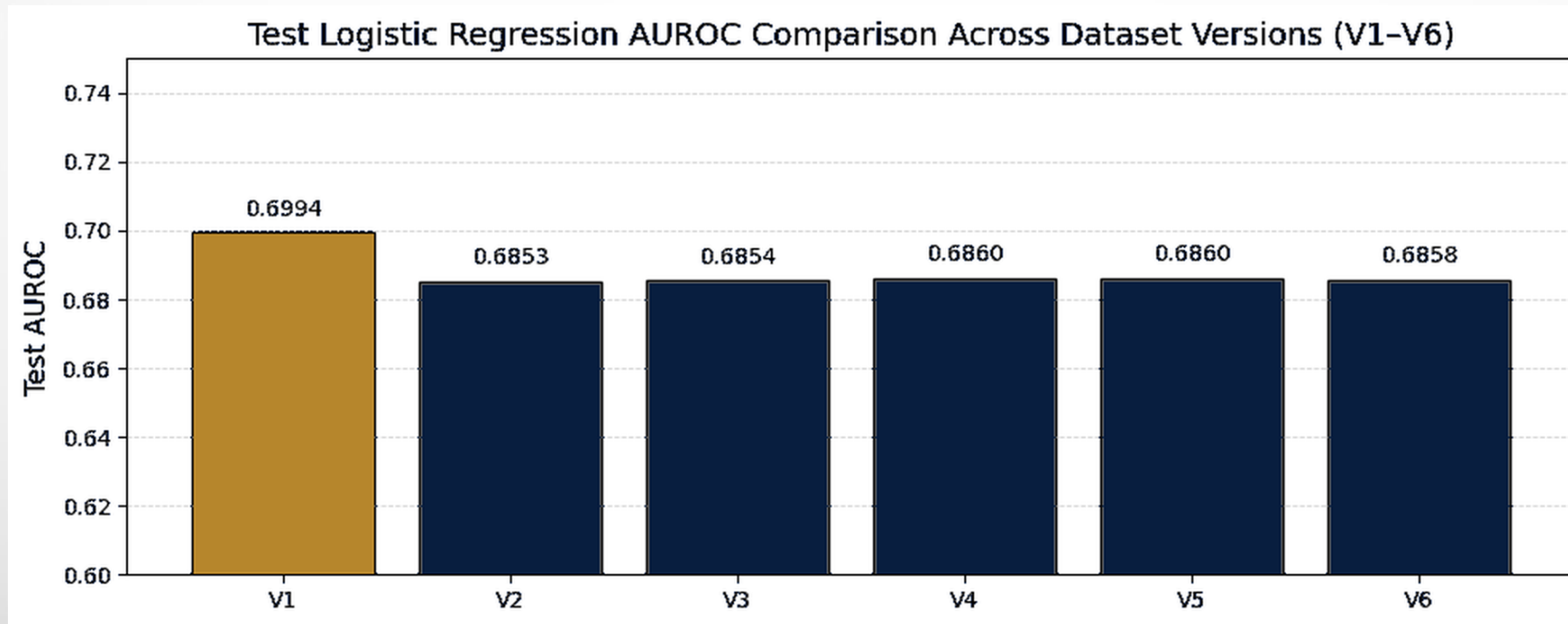


Model Performance

Logistic Regression Performance

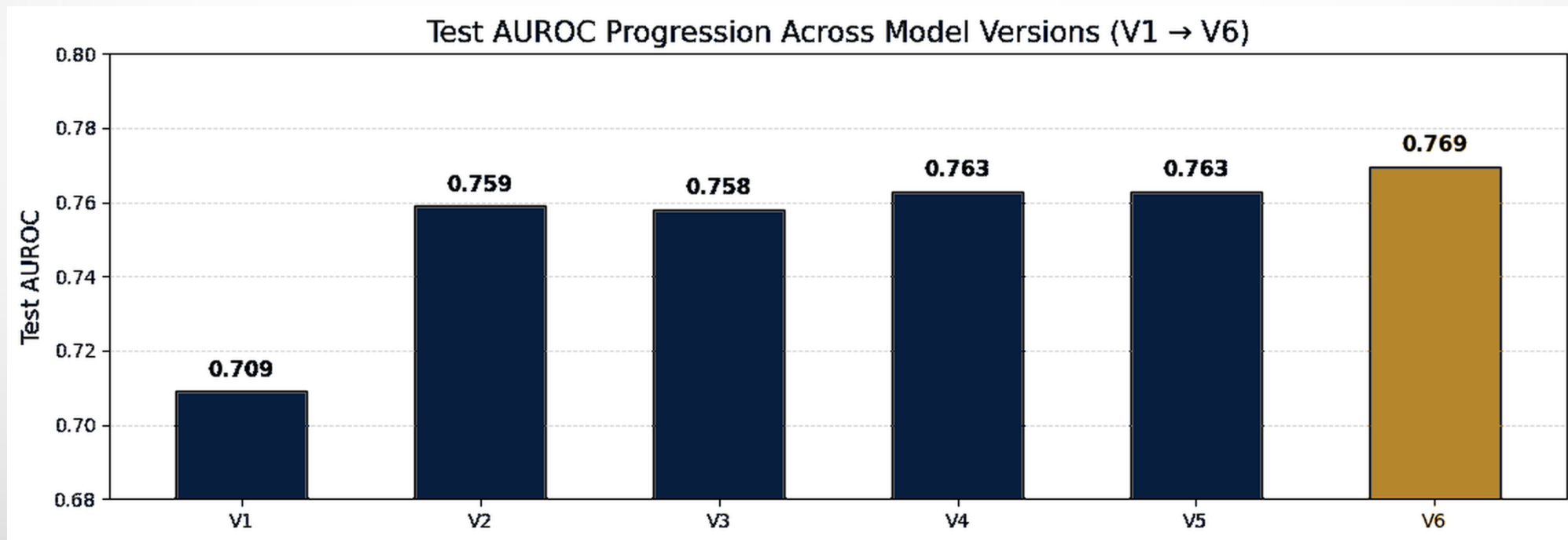
- **V1** of the Dataset gets the **best performance** from this baseline model

- With **increasing dataset complexity**, the logistic regression model starts to **underfit**.



LightGBM Model Performance

- **V1**: Baseline, lowest performance (0.709)
- **V2**: Biggest jump from added complexity
- **V3-V5**: No significant gains from additional features
- **V6**: Clinical counts boosted final AUROC (0.769)



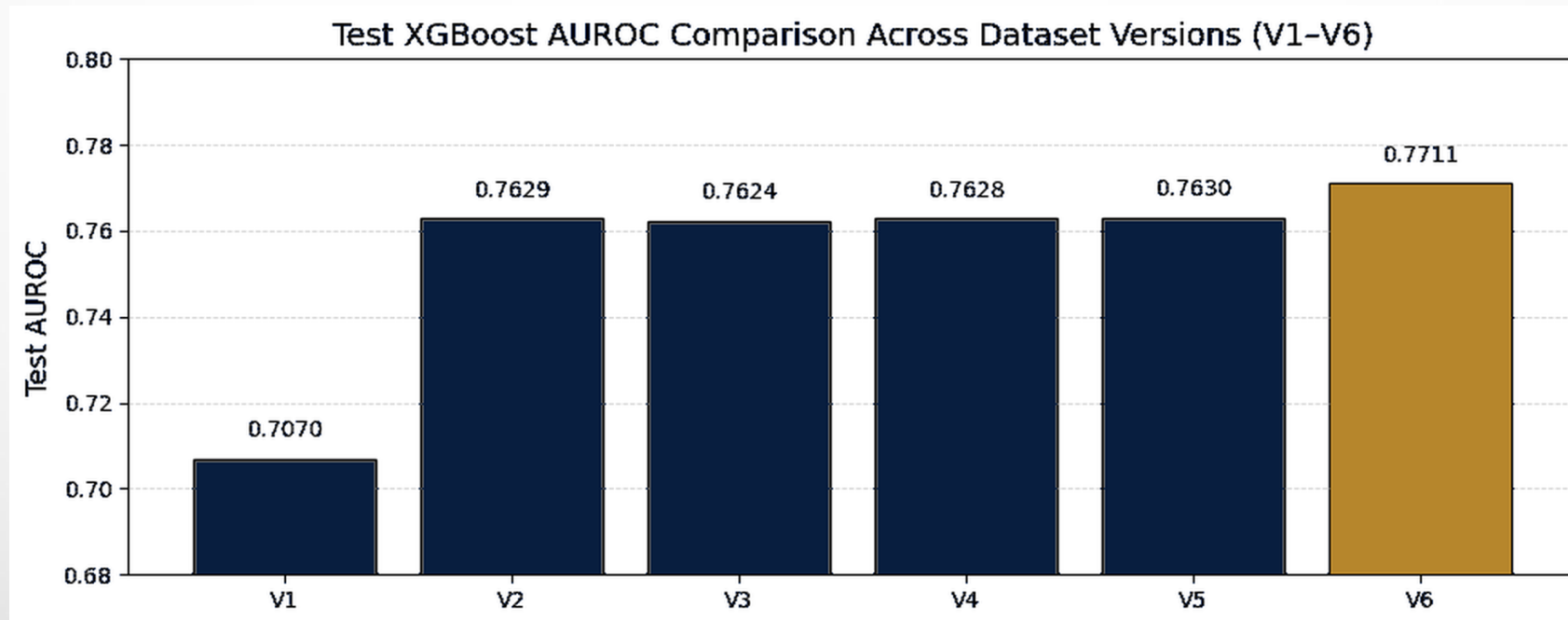
XGBoost Performance

- V1:** Baseline dataset with low features gave us low performance (0.7070)

- V2:** Biggest jump from added features and complexity (0.7629)

- V3-V5:** No Significant Gains

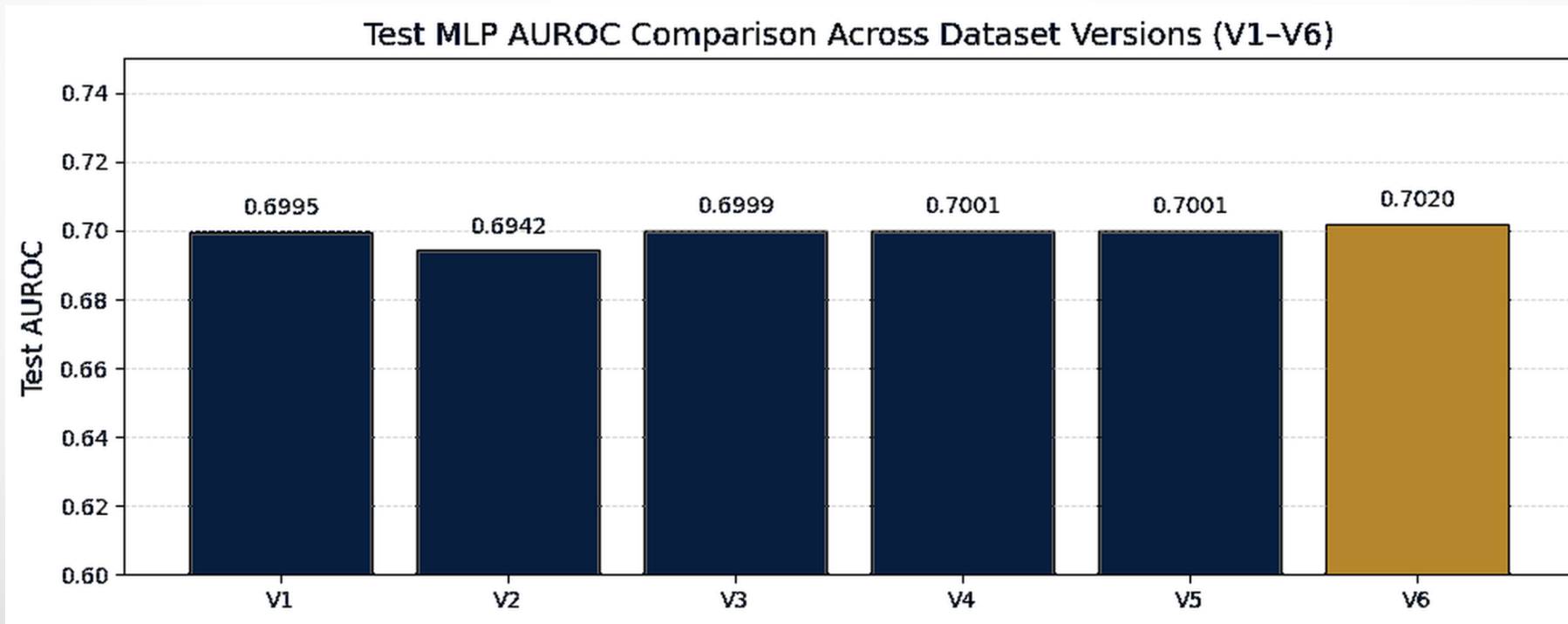
- V6:** Clinical counts features boosted final AUROC (0.7711)



MLP Performance

• **V6** obtains the best result but not by a significant margin.

• **Current Best Model Layer Architecture** of [128, 64] may be **underfitting** the complexity required from the datasets.



Advanced Neural Networks

Goal: Capture temporal dynamics, feature interactions and complementary signals beyond basic MLPs

Attention-Based tabular Model

FT-Transformer

Tokenizes each feature and uses self-attention to discover feature-to-feature interactions automatically.

Temporal Models

LSTM & GRU Hybrids

Sequence models that learn from patient admission history. Capture temporal patterns across prior hospitalizations.

Ensemble Stacking

Stacking Ensemble

Combines all architectures via a meta-learner that weights each model's predictions for optimal blending.

FT-Transformer

How It Works

1 Feature Tokenizer

Numeric + categorical features → individual d-dimensional vector

2 [CLS] Token

Learnable summary token prepended to the feature tokens

3 Self-Attention

Each token attends to all others: discovers which features interact

4 Classification

[CLS] output → LayerNorm → Dense(64) → Sigmoid → readmission probability

Why It Adds Value

MLPs

Blend all features together: interactions emerge implicitly through layers

LightGBM

Finds interactions via sequential tree splits: locally greedy

FT-Transformer

Explicitly learns which feature pairs matter via attention: global and differentiable

Different error patterns → valuable for stacking ensemble

Purpose-built Transformer architecture for tabular data (Gorishniy et al., 2021)

GRU+MLP Architecture

How the Hybrid Model Works

Sequence Branch

10 prior admissions × 15 temporal features → GRU layers → temporal representation

Static Branch

25 numeric features + 5 categorical embeddings → Dense layers

Fusion Layer

Concatenate both branches → Dense(128) → Dense(64) → Sigmoid

Table 2. Test AUROC

Model Family	Best Model	AUROC
Hybrid GRU	HybridGRU_1L32	0.7711
Hybrid LSTM	Hybrid_LSTM32	0.7695
Standalone RNN	BiGRU_2L64	0.6365

Key finding: Standalone RNNs (0.636) confirm that sequences alone are insufficient. The hybrid approach combining temporal + static features is essential.

Stacking Ensemble with Meta Learn

Layer 1: Base Models (trained independently)

Hybrid
GRU×4

MLP
×2

FT-Transformer
×3

LightGBM
×2

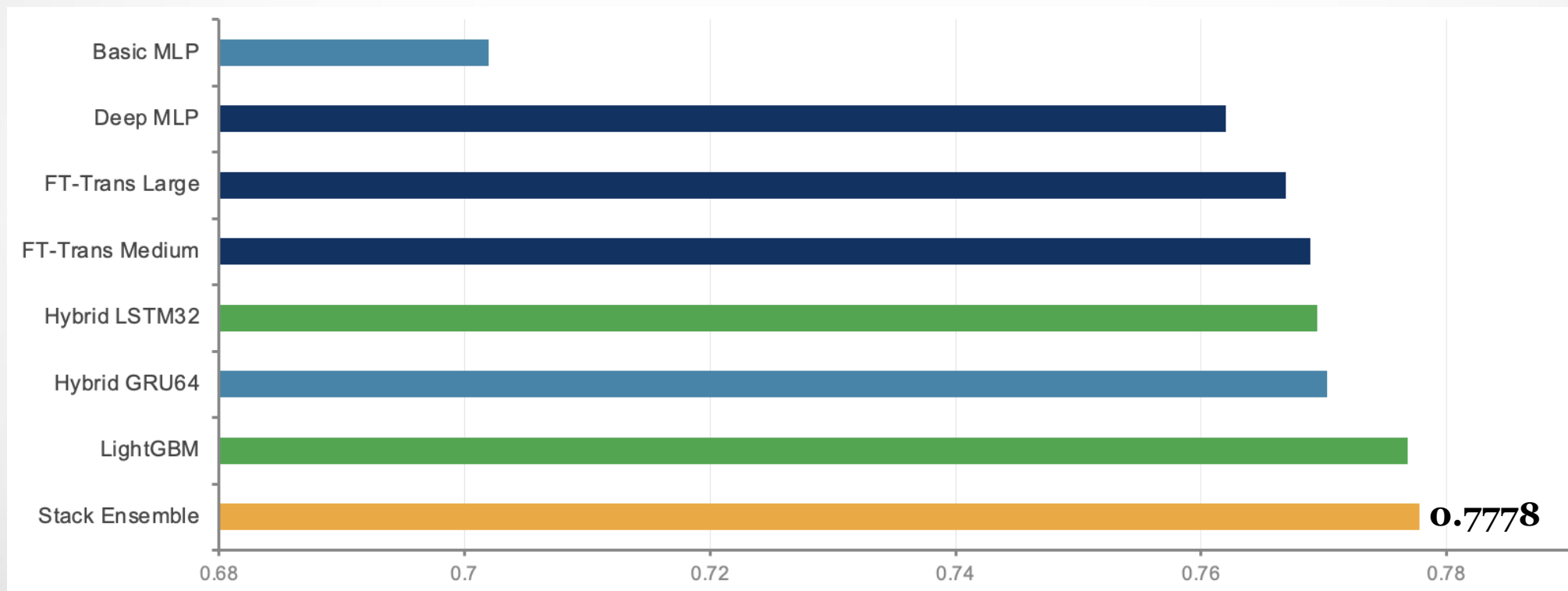
▼ Each model produces probability predictions ▼

Layer 2: Meta-Learner (Logistic Regression)

Learns optimal weights: which model to trust for which patients

▼
Final Prediction: AUROC = 0.7778

AUC-ROC Results on V6

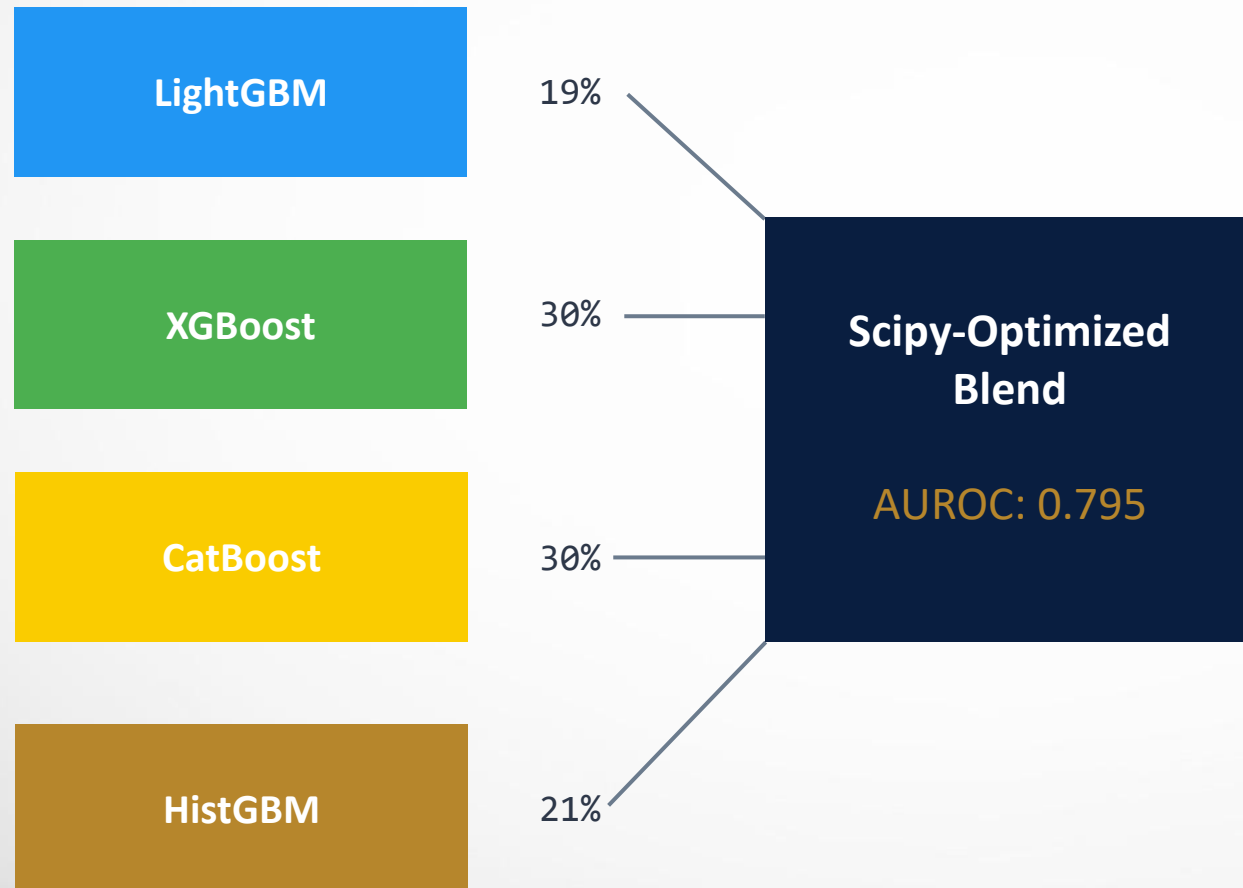


Test AUC-ROC across all model families and ensembles (V6 dataset, 244K admissions)

Boosting: Four-Model Ensemble with V7

LightGBM	XGBoost	CatBoost	HistGBM
0.790 AUROC	0.793 AUROC	0.792 AUROC	0.792 AUROC
Histogram-based, leaf-wise growth. Fast training, handles categoricals.	Lossguide tree policy Best individual performer	Ordered boosting Native categorical handling	Sklearn histogram gradient Native missing value support

Ensemble Blending Strategy



Blending Details

- Scipy minimize with AUROC objective on Out-of-Fold predictions
- Constrained: weights sum to 1.0, each weight in $[0, 1]$
- 10 random seeds per model for variance reduction
- Out-of-fold predictions prevent overfitting in meta-learner
- XGBoost and CatBoost receive highest weight (30% each)

Why Gradient Boosting for V7?



Temporal Features Not in V7

GRU/LSTM hybrids used 10 prior admissions $\times$ 15 temporal features. V7's new features are all static tabular, there is no sequences to leverage.



Diminishing Returns

Best ensemble (0.778) beat best single GBM (0.769) by only +0.009 AUROC. Neural nets did not outperform boosting on V6.



Interpretability Required

GBMs support full SHAP attribution which is essential for actionable discharge insights and answering RQ3.



V7: 50 Static Features

V6 (40 feat) + 5 target encodings + 5 clinical interactions = 50 features. All tabular $\rightarrow$ ideal for GBM architectures.

Conclusion: GBMs are the optimal architecture for V7 matching the data structure, preserving interpretability, and achieving 0.795 AUROC

Final Model Comparison:

V1 vs V6 vs V7 vs Expanded

Table 3. Final model comparison across dataset versions: Best AUROC, model type, interpretability, and deployment complexity.

Metric	V1 (21 feat)	V6 (40 feat)	V7 (50 feat)	Expanded
Best AUROC	0.709	0.778	0.795	0.800
Best Model	Logistic Reg.	Stacking	4-GBM Blend	4-GBM Blend
Interpretability	High	Moderate	High	Low
Deploy Complexity	Low	High	Low	Very High

V7 achieves 99.4% of the expanded model's performance with a compact, interpretable feature set. Additional feature engineering beyond the core clinical domains yielded only marginal improvement (+0.005 AUROC).

Benchmark Comparison

30-Day All-Cause Readmission Prediction on MIMIC Data

Published Benchmarks

Study	Approach	AUROC
LACE Index (van Walraven et al., 2010)	Traditional Clinical Score	0.684
Huang et al., 2020 (ClinicalBERT)	BERT + Clinical Notes	0.714
Ours (2026)	GBM Ensemble (50 features)	0.795

Key Takeaways

Outperforms Traditional Scores

Our model surpasses the LACE index by +0.111 AUROC, a 16% relative improvement

Exceeds NLP-Based Approaches

Achieves higher AUROC than ClinicalBERT without requiring clinical notes or NLP pipelines

Structured Data Only

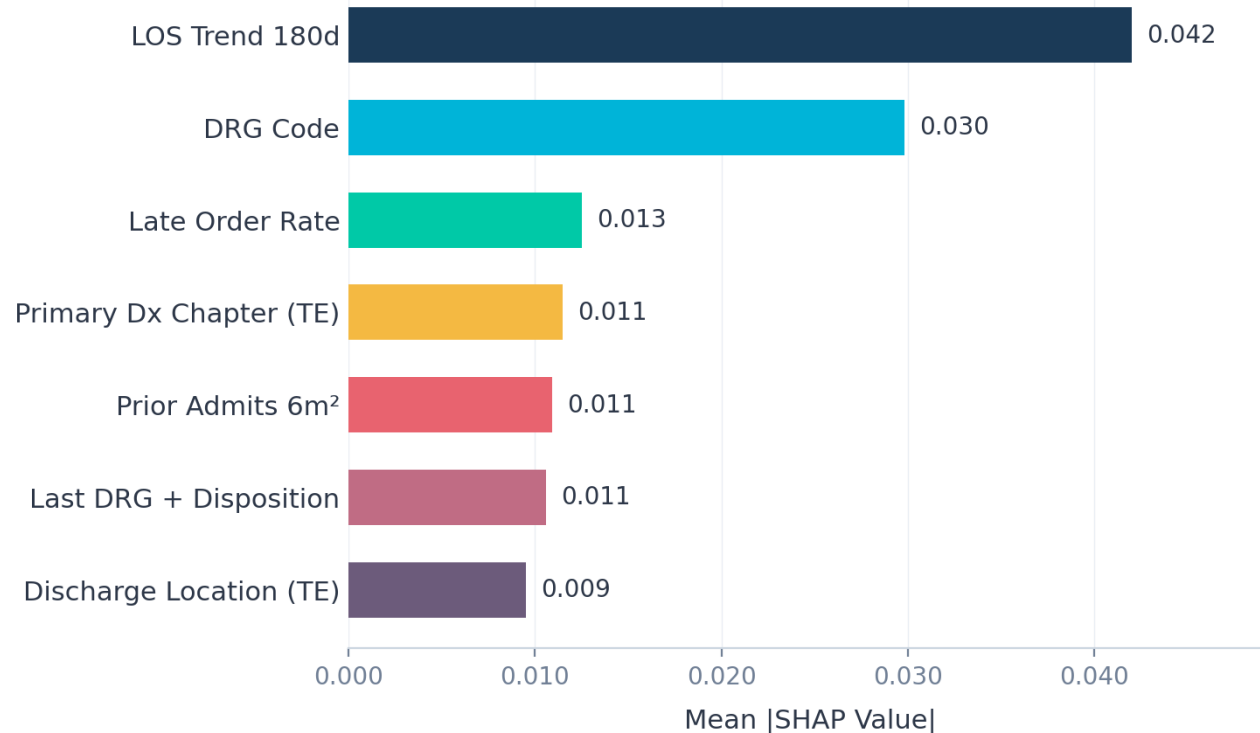
50 interpretable engineered features from structured EHR data — no notes, imaging, or temporal graphs

Clinically Deployable

Parsimonious feature set maps directly to actionable care interventions at discharge

SHAP: Top 7 Feature Explanations

Top 7 Feature Importance (SHAP)



Top 7 — Clinical Insights

- LOS Trend 180d — Rising stay lengths over 6 months are the strongest signal of instability
- DRG Code — Clinical mix carries enormous predictive power (11–66% readmission range)
- Late Order Rate — Operational chaos during stay predicts post-discharge failure
- Primary Dx Chapter — Disease category shapes baseline risk profile
- Prior Admits 6m² — Acceleration in utilization compounds risk non-linearly
- Last DRG + Disposition — Prior discharge pathway predicts future readmission patterns
- Discharge Location — Where patients go after discharge directly impacts readmission risk

Answering the Research Questions

RQ1**Most Predictive Features**

LOS trend (180d), DRG code, discharge location, prior admission history, and clinical interactions are the top predictors. A core set of 50 features captures 99.4% of predictive performance.

RQ2**Best Modeling Approach**

XGBoost (AUROC: 0.793) was selected over the top-performing blended ensemble (0.795), as the negligible 0.002 difference does not justify the added complexity and XGBoost preserves full feature interpretability for actionable insights.

RQ3**Actionable SHAP Insights**

SHAP provides patient-level explanations: providers see which factors drive each individual's risk. Key: LOS monitoring, medication reconciliation, early follow-up for frequent admitters.

Limitations

Single Hospital System

Data from Beth Israel Deaconess (Boston) only, which may limit generalizability to other hospitals, regions, or populations.

No Unstructured Clinical Notes

Physician notes, discharge summaries, and radiology reports were not used. NLP-based features could improve predictions.

No Social Determinants of Health

Housing stability, social support, health literacy, and transportation are known risk factors but unavailable in MIMIC-IV.

Diminishing Returns

Expanding features beyond the core clinical set yielded only marginal improvement, suggesting a ceiling with available structured data.

Temporal Limitation

30-day readmissions occurring at different hospitals are not captured in MIMIC-IV.

Future Work

External Validation

Test on data from other hospital systems to assess generalizability.

NLP Integration

Incorporate unstructured clinical notes using text mining for richer patient context.

Real-Time Deployment

Develop an API/dashboard for real-time risk scoring within EHR workflows at discharge.

Fairness Analysis

Evaluate performance across demographic subgroups to identify and mitigate biases.

Causal Inference

Move beyond prediction to identify causal factors enabling more effective interventions.

Cost-Effectiveness

Quantify the economic impact of implementing the prediction model in hospital operations.

Questions or Comments

Thank you!